

Course Name:High Performance Computing			
Course Code:		Course Credits:	3
Teaching Hours / Week (L: T: P):	3	CA Marks:	40
Total Number of Teaching Hours:	45	END-SEM Marks:	60
Course Prerequisites: Edge Computing ,Cloud Computing			
Course Description: Explore the fundamentals of High Performance Computing (HPC) in this course, covering parallel computing architectures, algorithms, and software tools for solving computationally intensive problems efficiently. Gain insights into optimizing code for parallel execution on multi-core processors, clusters, and supercomputers. Topics include message passing interfaces, distributed memory programming, GPU computing, and performance analysis techniques. Hands-on labs provide practical experience in HPC application development, performance tuning, and scalability assessment. By the end, students will be equipped with essential skills for harnessing the power of modern computing architectures to tackle complex scientific, engineering, and data-intensive challenges			
Course Objectives: (3 to 5) The main objectives of the course are : 1.To provide students with a comprehensive understanding of parallel computing, parallel programming platforms, and associated concepts. 2.To equip students with a strong foundation in the principles of parallel algorithm design. 3.To provide students with a comprehensive understanding of various parallel communication operations commonly used in parallel and distributed computing environments. 4.To cater students with a deep understanding of the theoretical and practical aspects of parallel programming, focusing on identifying and mitigating sources of overhead. 5.To equip students with a comprehensive understanding of the challenges and strategies involved in parallel sorting algorithms and graph algorithms.			
Course Outcomes: (Five) – Use Bloom's Taxonomy On completion of the course, learner will be able to– 1. Understand fundamental concepts in High-Performance Computing (HPC) and parallel computing architectures. 2. Develop proficiency in designing and implementing parallel algorithms for various parallel computing architectures. 3. Demonstrate proficiency in performance optimization techniques for parallel computing systems. 4. Foster collaboration and teamwork skills in HPC projects 5. Apply ethical principles in the development and use of HPC solutions, considering societal implications and ethical considerations.			
Unit I	Background and Introduction to HPC		(9 Hours)

Introduction to Linux, Basics of Computer Architectures, Flynn's Taxonomy for computer architectures, Linux Commands, Processes scheduling & synchronization, Memory management, File Systems, Shell Scripting, Package Managers, Basics of C, Fundamentals, Basic Syntax and Structures, Control Flow, Functions, Arrays etc. Introduction to Supercomputer and components, Important factors in supercomputer, Performance Metrics, Parallel Programming Overview, Parallel architecture, Memory architectures, Parallel programming paradigms, HPC system access and environment setup, HPC job submission, Overview of SLURM, commands, and usage, Submitting jobs to HPC system.

Pedagogy	ICT Teaching / PowerPoint Presentation and Videos on- parallel Architectures
	Self-study / Do it yourself:
	Experiential Learning Topics:
	Case Study / PBL - Project Based Learning:

Unit II	Shared Memory Programming Model	(9 Hours)
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OpenMP memory model, OpenMP directive, Runtime Library and environment variables, Special features of OpenMP, Critical section, Atomic and reduction, Variable scoping etc. OpenMP examples and application, Compilation Debugging and bug fixing, OpenMP assignments Case Study: Matrix Multiplication

Pedagogy	ICT Teaching / Power Point Presentation and Videos
	Self-study / Do it yourself:
	Experiential Learning Topics:
	Case Study / PBL - Project Based Learning: A Brief History of GPU's

Unit III	Distributed Memory Programming Model	(9 Hours)
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Parallel Programming models, Introduction to MPI, Point to point communication calls, Collective Communication calls, Demonstration of Sample MPI programs etc. Case Study: Trapezoidal Rule

Pedagogy	ICT Teaching / Power Point Presentation and Videos
	Self-study / Do it yourself:
	Experiential Learning Topics: One-to-All Broadcast and All-to-One Reduction,
	Case Study / PBL - Project Based Learning:

Unit IV	Accelerated Computing – GPU's	(9 Hours)
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Introduction to GPU Architectures, Basics of CUDA Programming, CUDA Process flow and Data transfer, CUDA parallel threads, Streaming Multi-processors (SM's), Threads, blocks, Grids etc. Case study: Matrix – Vector Multiplication

Pedagogy	ICT Teaching / Power Point Presentation and Videos
	Self-study / Do it yourself:
	Experiential Learning Topics:
	Case Study / PBL - Project Based Learning:

Unit V	Profiling and Debugging of Parallel Programs	(Hours)
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Introduction to profiling, Debugging of Parallel codes, Optimization of Parallel codes, Optimization techniques, Compiler optimization, library optimization, source code optimization, Basics of profiling tools gprof, Intel Vtune etc. HPC Applications – Use cases (Exploration of HPC use cases for different scientific and engineering domains, HPC applications installations and environment setup etc, Case study: e.g NAMD)

Pedagogy	ICT Teaching / Power Point Presentation and Videos
	Self-study / Do it yourself: Parallel DFS
	Experiential Learning Topics:
	Case Study / PBL - Project Based Learning:

Learning Resources

Text Books:

1. Peter S. Pacheco, An Introduction to Parallel Programming, Elsevier, First Edition (1 January 2011)
2. David B. Kirk, Programming Massively Parallel Processors: A Hands-on Approach, Morgan Kaufmann, 4th edition (23 September 2022)

Reference Books:

1. Kai Hwang, "Scalable Parallel Computing", McGraw Hill 1998, ISBN:0070317984
2. David Culler Jaswinder Pal Singh, "Parallel Computer Architecture: A Hardware/Software Approach", Morgan Kaufmann, 1999, ISBN 978-1-55860-343-1
3. Rod Stephens, "Essential Algorithms", Wiley, ISBN: 978-1-118-61210-1

MOOC/ Video Lectures available at:

ASSESSMENT AND EVALUATION PATTERN		
WEIGHTAGE	Continuous Assessment (CA)	End Semester Assessment (ESA)
CA - Assignments (PBL / Case Study / Presentation / Seminar / Group Discussions / Quiz / Test		
ESA - Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating		

Project Based Learning – Mini (Minor) Project / Major Project

COs	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	Relevant to National/ Global/Regional
CO-1	3	3	3		3	3		3	3			2	2		Global
CO-2	3	3	3	3								2			Global
CO-3	3	3	3										2		Global
CO-4						3	3		3			2	2		Global
CO-5				3	3				3	3					Global

3 – HIGH, 2 – MEDIUM, 1 - LOW